

SOFTWARE ENGINEERING

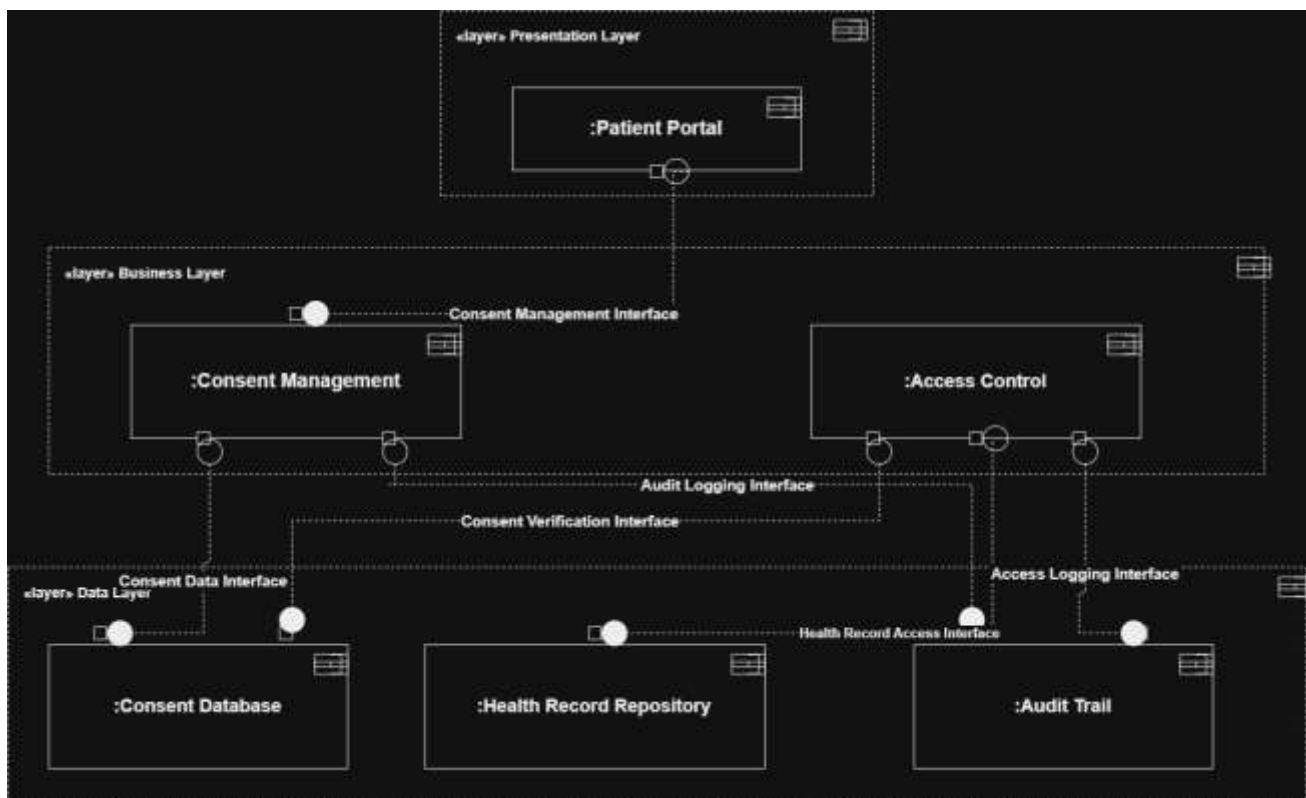
LAB 3 – ARCHITECTURAL JUSTIFICATION

Patient Health Record Consent Management System

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UML DIAGRAM



REPORT

1. Architectural Choice

For the Patient Health Record Consent Management System, Layered Architecture has been selected as the guiding architectural style. The system is organized into three distinct layers, each responsible for a well-defined aspect of functionality. The Presentation Layer consists of the Patient Portal, which manages all direct interaction with patients and handles the presentation of information and consent-related actions. The Business Layer contains the Consent Management and Access Control components, which together implement the core logic governing how consent is granted, verified, and enforced, and how access to sensitive health data is authorized. The Data Layer comprises the Consent Database, the Health Record Repository, and the Audit Trail, and is responsible for the persistent storage and retrieval of consent records, patient health data, and a verifiable log of system activity. This layered separation ensures that each concern of the system is isolated and handled by the appropriate component.

2. Reason 1 – Separation of Concerns

Layered Architecture is well suited to this system because it enforces a strict separation of concerns across presentation, business logic, and data management. The Patient Portal is concerned only with user interaction and

does not contain any logic for validating consent or enforcing access rules; that responsibility belongs entirely to the Business Layer. Similarly, the Consent Management and Access Control components operate independently of how data is physically stored, since persistence is delegated to the Data Layer. This clear division prevents responsibilities from becoming mixed, so that changes to how patients interact with the portal, how consent decisions are made, or how records are stored can be reasoned about and modified without unintended interference between layers.

3. Reason 2 – Maintainability and Modularity

Placing Consent Management, Access Control, and the data components into their respective layers makes the system considerably easier to maintain, test, and extend over time. For example, if a new consent policy or an additional access rule needs to be introduced, the change can be made within the Business Layer without requiring modification of the Patient Portal or the underlying data stores. Likewise, the Consent Database, Health Record Repository, and Audit Trail can each evolve or be optimized independently, as long as the interfaces they expose to the Business Layer remain consistent. This modularity also simplifies testing, since each layer can be validated in isolation before integration.

4. Security Advantage

A significant security advantage of Layered Architecture in this system is that it centralizes and isolates access control and consent verification within the Business Layer, rather than allowing the Presentation Layer or Data Layer to make authorization decisions independently. Every request originating from the Patient Portal must pass through the Access Control and Consent Management components before any health record can be retrieved from the Health Record Repository. This ensures that access to sensitive patient health information is always mediated by consistent, centrally enforced rules, and that the Data Layer never exposes records directly to the presentation logic. Combined with the Audit Trail recording access and consent events, this structure reduces the risk of unauthorized or unverified access to sensitive health data.

5. Performance Benefit

A realistic performance benefit of this layered structure is that responsibilities can be optimized independently within each layer without affecting the others. Because consent verification and access checks are handled entirely within the Business Layer, the Data Layer is only invoked once a request has already been validated, avoiding unnecessary processing or unauthorized queries against the Consent Database or Health Record Repository. This allows components such as the Audit Trail or Health Record Repository to be tuned or scaled according to their own load characteristics, without requiring changes to the Patient Portal or the business logic. It should be noted that this benefit arises from targeted optimization opportunities rather than any inherent speed advantage of layering itself.

6. Overall Architectural Justification

Taken together, Layered Architecture is an appropriate choice for the Patient Health Record Consent Management System because it provides a clear separation of responsibilities between patient interaction, consent and access logic, and data storage. This separation strengthens the security of sensitive health information by ensuring that access and consent are consistently verified in one place, supports maintainability by allowing individual layers to change independently, and results in organized, well-structured management of consent records, health data, and audit information. For these reasons, Layered Architecture aligns closely with the functional and non-functional requirements of this system.